

Chun-Ju Lai

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Academia Sinica

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Appointments

2020 – now	Assistant Research Fellow, Academia Sinica
2017 – 2020	Limited Term Assistant Professor, University of Georgia , USA
2016 – 2017	Postdoc Fellow, Max-Planck Institute for Mathematics in Bonn , Germany

Experience

2022	Recipient of the 2021 Young Scientist Award, National Center of Theoretical Sciences
2021 – 2025	Program Committee Member, National Center of Theoretical Sciences , Taiwan
2019	Summer Collaborator Program, Institute for Advanced Study , USA

Education

2011 – 2016	Ph.D. Mathematics, University of Virginia , USA
2010 – 2011	Research Assistant, National Center of Theoretical Sciences , Taiwan
2005 – 2009	B.S. Mathematics; B.S.E. Electrical Engineering, National Taiwan University , Taiwan

Publications

- [1] S. Eu, T. Fu, and C. Lai, *On the enumeration of parking functions by leading terms*, Adv. in Appl. Math., 35 (2005), 392–406.
- [2] S. Eu, T. Fu and C. Lai. *Cycle lemma, parking functions and related multigraphs*, Graphs Combin., 26 (2010), 345–360.
- [3] C. Lai, *On Weyl modules over affine Lie algebras in prime characteristic*, Transform. Groups 21 (2016), 1123–1153.
- [4] C. Lai and L. Luo, *An elementary construction of monomial bases of modified quantum affine $\mathfrak{gl}_n$* , J. Lond. Math. Soc. 96 (2017), 15–27.
- [5] Z. Fan, C. Lai, Y. Li, L. Luo, and W. Wang, *Affine flag varieties and quantum symmetric pairs*, Mem. Amer. Math. Soc., 265 (2020), no. 1285.
- [6] Z. Fan, C. Lai, Y. Li, L. Luo, W. Wang and H. Watanabe, *Quantum Schur duality of affine type C with three parameters*, Math. Res. Letters, 27 (2020) 79–114.
- [7] C. Lai and L. Luo, *Schur algebras and quantum symmetric pairs with unequal parameters*, Int. Math. Res. Not., 13 (2021) 10207–10259.
- [8] C. Lai, D. K. Nakano and Z. Xiang, *On Schur algebras corresponding to the Hecke algebras of type B* , Transform. Groups, 27 (2022) 983–1024.
- [9] Z. Fan, C. Lai, Y. Li, L. Luo, and W. Wang, *Affine Hecke algebras and quantum symmetric pairs*, Mem. Amer. Math. Soc. , 281(2023) no. 1386.
- [10] M.S. Im, C. Lai and A. Wilbert, *A study of irreducible components of Springer fibers using quiver varieties*, J. Algebra, 591 (2022) 217–248.
- [11] M.S. Im, C. Lai and A. Wilbert, *Irreducible components of two-row Springer fibers for all classical types*, Proc. Amer. Math. Soc., 150 (2022), 2415–2432.
- [12] C. Lai, D. K. Nakano and Z. Xiang, *Quantum wreath products and Schur-Weyl duality I*, Forum Math. Sigma, 12 (2024): e108.
- [13] Y. Hsu and C. Lai, *On the Springer correspondence for wreath products*, J. Pure Appl. Algebra, 230 (2026), 108272.
- [14] C. Lai and A. Minets, *Schurification of polynomial quantum wreath products*, Represent. Theory, to appear.
- [15] C. Lai, D. K. Nakano and A. Wilbert, *Category $\mathcal{O}$ for Lie superalgebras*, arXiv:2505.17422, 26 pages.
- [16] C. Lai and C. Li, *Infinite-dimensional towers and a categorification of differential operators on $\mathbb{A}_{\mathbb{Z}[v, v^{-1}]}^1$* , arXiv:2509.20053, 10 pages.

- [17] C. Lai, D. K. Nakano and Z. Xiang, *Quantum wreath products and Schur-Weyl duality II*, arXiv:2511.19825, 44 pages.
- [18] V. Buciumas and C. Lai, *Quantum wreath products and p -adic general linear group*, arXiv:2602.20817, 28 pages.
- [19] B. Hudak and C. Lai, *On Cellularity of Hecke Algebras for Wreath Products*, arXiv:2606.03759, 15 pages.

Grants

- **National Science and Technology Council (NSTC) Grant:**
 - The 2030 Cross-generation Emerging Young Scholars, (2022–2026).
 - NCTS operational project, co-PI, (2023–2026).
- **Ministry of Science and Technology (MoST) Grant:**
 - Research Project Grant, (2020–2023).
 - NCTS operational project, co-PI, (2021–2023).
- **Provost’s Affordable Course Materials Grant**, University of Georgia, with J. Hoisington, (2019–2020).
- **ICM 2022 Kovalevskaya Grant** for travel reimbursement.

Organizations

1. Conferences:

- **Pacific Rim Conference on Mathematics**, Academia Sinica, Taiwan, Jun 15–19, 2026.
- **Conference on Algebraic Representation Theory (CART)**, NCTS, Taiwan, Dec 1–3, 2023.
- **AMS Special Sessions**, University of Virginia, Mar 11–13, 2022; Georgia Tech, Mar 17–18, 2023.
- **TMS Annual Meeting**, Academia Sinica, Jan 17–18, 2022.
- **AMS Special Session on Representation Theory of Algebraic Groups and Quantum Groups: A Tribute to the Work of Cline, Parshall and Scott (CPS)**, University of Virginia, Mar 11–13, 2022.
- **Retirement Conference of Yeong-Nan Yeh**, Academia Sinica, Jun 27–28, 2021.
- **Southeast Lie Theory Conference XI**, University of Georgia, Jun 10–12, 2018.

2. Seminars:

- **Colloquium at the Academia Sinica**, 2020 - now.
- **Learning Seminar on Representation Theory**, Academia Sinica, 2020 - 2023.
 - 2023 Fall: on R -matrices
 - 2023 Spring: on Integral Hecke Algebras
 - 2022 Spring: on Double Affine Hecke Algebras
 - 2021 Fall: on Geometric representation theory
 - 2021 Spring: on Kac-Moody groups
 - 2020 Fall: on Soergel bimodules
- **Basic Notion Seminar**, Academia Sinica/National Taiwan University, 2020 - 2021.
- **Algebra Seminar at the University of Georgia**, 2018 – 2020.

Training and Service

• Journal Manuscript Reviewer:

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| <ul style="list-style-type: none"> – <i>Advances in Mathematics</i> – <i>Algebra Colloquium</i> – <i>Algebras and Representation Theory</i> – <i>Annals of Representation Theory</i> – <i>Communications in Algebra</i> – <i>European Journal of Combinatorics</i> – <i>Formal Power Series and Algebraic Combinatorics</i> – <i>International Mathematics Research Notices</i> – <i>Journal of Algebra</i> | <ul style="list-style-type: none"> – <i>Journal of Algebra and its Applications</i> – <i>Journal of Pure and Applied Algebra</i> – <i>Letters in Mathematical Physics</i> – <i>Mathematical Research Letters</i> – <i>Nagoya Mathematical Journal</i> – <i>Pure and Applied Mathematics Quarterly</i> – <i>Quarterly Journal of Mathematics</i> – <i>Representation Theory</i> – <i>Symmetry</i> – <i>Transformation Groups</i>. |
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- **Supervising:**
Postdocs:
 - Akito Uruno (NCTS, 2026–)
 - Berta Hudka (NCTS, 2024–2026; current: industry)
 - Cailan Li (AS, 2024–)
 - You Hung Hsu (NCTS, 2022–2024; current: NCKU faculty)
 PhD students:
 - Ziqing Xiang (Univ. Georgia; current: SUSTech faculty)
 Research Assistants:
 - Wei-Jiun Kao (NCTS, 2022–2023; next: PhD student at Univ. Michigan)
 - Cheng-Jui Yeh (NCTS 2023–2024; next: MS at Univ. Minnesota Twin Cities)
 Master students:
 - Ming-Jun Lin (NTHU, 2024; current: industry).
- **Thesis Committee Examiner:**
Ph.D. Examiner:
 - Aayush Bhattacharya, The University of New South Wales Sydney, 2026
 Master's Thesis Examiner:
 - Wen-Yen Cheng, National Taiwan University, 2026.
 - Jhih-You Syu, National Tsing Hua University, 2025
 - Yu-Lun Cheng, National Taiwan University, 2022.
- **Science Fair Judge:**
 - of the Georgia Science and Engineering Fair (GSEF), 2018 – 2020.
 - of the Shing-Tung Yau High School Mathematics Award 2020–2021.
 - of the Taiwan International Science Fair 2023.
 - of the National Primary and High School Science Fair 2025.
- **Science communicator**
 - for the *Academia Sinica Newsletter*, 2020, regarding representation theory.
Editor of *Academia Sinica Newsletter* 2020–2022.
 - for the *Math Media* Vol. 176, 2020.
 - for the *Taiwan Top Science Student Project*, 2022.

Conference Talks

2007	Jun	Cross-strait Conference on Graph Theory and Combinatorics, National Taiwan University, Taiwan
2014	May	Southeastern Lie Theory Workshop, University of Georgia, USA
2015	Mar	AMS special session on <i>Quantum Algebras, Representations, and Categorifications</i> , Georgetown University, USA
	Jul	Workshop on Lie Theory and Representation Theory, East China Normal University, China
	Oct	Southeastern Lie Theory Workshop, North Carolina State University, USA
2016	May	Workshop on Hecke Algebras and Lie Theory, University of Ottawa, Canada
	May	Southeastern Lie Theory Workshop, University of Virginia, USA
2017	Jul	Taipei Workshop on Representation Theory of Lie Superalgebras and Related Topics, Academia Sinica, Taiwan
	Sep	AMS special session on <i>Recent Advancements in Representation Theory</i> , University at Buffalo (SUNY), USA
2018	Oct	Representation Theory, Combinatorics, and Geometry, University of Virginia, USA
2019	Mar	AMS special session on <i>Representations of Lie algebras, algebraic groups, and quantum groups</i> , Auburn University, USA
	May	Southeastern Lie Theory Workshop, Louisiana State University
	Nov	AMS special session on <i>Geometric Methods in Representation Theory</i> , University of California Riverside, USA
	Nov	AMS special session on <i>Geometry and Representation Theory of Quantum Algebras and Related Topics</i> , University of California Riverside, USA
2020	Aug	Taipei workshop on representation theory and related topics, Academia Sinica, Taiwan

	Oct	AMS special session on Geometry and Representation Theory of Quantum Algebras and Related Topics, University of Utah, USA
	Dec	TMS Annual Meeting, Fu-Jen Catholic University, Taiwan
2021	Nov	Conference on Algebraic Representation Theory (CART), Seoul National University, Korea
2022	Nov	Two-day Workshop on Lie Theory, Fudan University, China
2023	Jan	TMS Annual Meeting, National Tsing Hua University, Taiwan
	Jun	Representation Theory of Hecke Algebras and Categorification, OIST, Japan
	Aug	International Algebra Conference in the Philippines (IACP), Philippines
2024	Jan	TMS Annual Meeting, National Cheng Chi University, Taiwan
	Mar	One Day Workshop in Lie Theory, Beijing Institute of Technology, Beijing, China.
	May	AMS special session on Diagrammatic Algebras in Representation Theory and Beyond, San Francisco State University, USA
	May	Representation Theory and Related Geometry: Progress and Prospects, University of Georgia, USA
	Aug	Lie Theory and Related Topics, Sichuan University, Chengdu, China.
2025	Feb	Lie Theory and Related Topics, New York University Abu Dhabi, UAE.
	Feb	NCTS Workshop on Lie Theory and Representation Theory, Taiwan
	May	International Workshop on Lie Algebras, Shenzhen, China
	Jun	9th International Conference on Representation Theory, Nanjing, China
	Oct	KMS Special Session, Seoul, South Korea
2026	Jan	Winter Mathematics Research International Workshop – Lie Theory and Related Areas, Muscat, Oman
	Mar	Recent Developments on Quantum Groups and Related Topics, Hong Kong, China
	Mar	AMS special session on “Algebraic, combinatorial and geometric aspects of representation theory”, Georgia Southern University, USA
	Jul	ICM 2026 Satellite conference, University of Virginia, USA
	Oct	Group representation theory and related topics, Tianyuan Mathematics Research Center, China

Expository Talks

2014	Jul	5 lectures, <i>Minicourse on Representation Theory of Lie Algebras</i> , Taiwan Normal University, Taiwan
	Dec	4 lectures, <i>Minicourse on Representation Theory of Lie Algebras</i> , Central University, Taiwan
2016	Jul	3 lectures, <i>Minicourse on Quantum Groups</i> , Taiwan Normal University, Taiwan
2020	Oct	10 lectures, <i>Minicourse on Springer Fibers and Quiver Varieties</i> , National Center of Theoretical Science, Taiwan
2023	Aug	2 lectures, <i>Minicourse on Quantum Wreath Products</i> , National Center of Theoretical Science, Taiwan
2026	Oct	4 lectures, <i>Quantum Wreath Products and Related Conjectures</i> , Tianyuan Mathematics Research Center, China

Seminar Talks

2010	Nov	Combinatorics Seminar, University of Kaohsiung, Taiwan
2013	Sep	Algebra seminar, University of Virginia, USA
	Nov	Graduate seminar, University of Virginia, USA
2014	May	Algebra seminar, University of Virginia, USA
	Dec	Seminar, Central University, Taiwan
	Dec	Seminar, Academia Sinica, Taiwan
2015	Nov	Algebra seminar, University of Virginia, USA
	Dec	Lie Group/Representation Theory Seminar, University of Maryland, USA
2016	Sep	Algebra Seminar, University at Buffalo (SUNY), USA
	Oct	Emmy Noether Seminar, FAU Erlangen-Nürnberg, Germany
	Nov	Algebra Seminar, Newcastle University, UK
	Dec	Oberseminar, Max Planck Institute for Mathematics in Bonn, Germany
	Dec	Algebra Seminar, Universität Bonn, Germany

	Dec	Algebra Seminar, Institut de mathématiques de Jussieu-Paris Rive Gauche, France
2017	Feb	Algebra Seminar, Université Clermont Auvergne, France
	Apr	Algebra Seminar, Universität Bielefeld, Germany
	Aug	Algebra Seminar, University of Georgia, USA
	Aug	Oberseminar, University of Georgia, USA
	Sep	Algebra Seminar, University of Georgia, USA
2018	Apr	Algebra Seminar, University of Georgia, USA
	Sep	Algebra Seminar, University of Georgia, USA
	Dec	Combinatorics Seminar, Academia Sinica, Taiwan
2019	Jan	Algebra Seminar, University of Georgia, USA
	Sep	Algebra Seminar, University of Georgia, USA
	Dec	Combinatorics Seminar, Sun Yat-Sen University, Taiwan
	Dec	Combinatorics Seminar, Academia Sinica, Taiwan
	Dec	Colloquium, Tsing Hua University, Taiwan
2020	Feb	Algebra Seminar, Taiwan University, Taiwan
	Nov	Basic Notions Seminar, Taiwan University, Taiwan
	Nov	Algebra Seminar, Jagiellonian University, Poland
	Dec	Algebra Seminar, Cheng Kung University, Taiwan
2021	Jan	Colloquium, Central University, Taiwan
	May	Colloquium, Tsing Hua University, Taiwan
	Sep	QSMS Seminar, Center for Quantum Structures in Modules and Spaces, Korea
	Oct	Colloquium, Chengchi University, Taiwan
2022	Mar	Algebra Seminar, Cheng Kung University, Taiwan
	Feb	Colloquium, Taiwan University, Taiwan
	Mar	Algebra Seminar, University of Georgia, USA
	Mar	Algebra Seminar, University of Virginia, USA
	Mar	Awardee Lecture, NCTS Spring Day, NCTS, Taiwan
	Sep	Algebra Seminar, Uppsala University, Sweden
	Oct	"What are you thinking about?" Seminar, University of Amsterdam, Netherlands
	Oct	Algebra Seminar, University of Cologne, Germany
	Nov	Representation Theory Seminar, OIST, Japan
2023	Mar	TACO Seminar, Loyola University Chicago, USA
	Mar	Algebra Seminar, California State University Northridge, USA
	Mar	Algebra Seminar, University of Virginia, USA
	Mar	Algebra Seminar, University of Georgia, USA
	Oct	Colloquium, Central University, Taiwan
	Nov	Algebra Seminar, SUSTech, China
2024	Mar	Algebra Seminar, Peking University, China
	Apr	Student Geometry Seminar, Tsing Hua University, Taiwan
	Apr	Algebra Seminar, University of Oregon, USA
	Oct	Algebra Seminar, Universität Bonn, Germany
2025	Mar	London Algebra Colloquium, London, UK
	May	Algebra Seminar, Harbin Institute University, China
	Sep	Representation Theory Seminar, Pohang Science and Technology University, South Korea
	Oct	Representation Theory Seminar, Seoul National University, South Korea
2026	Feb	Representation Theory Seminar, University of Hong Kong, Hong Kong
	Mar	Colloquium, University of South Alabama, USA
	Apr	Algebra Seminar, University of Georgia, USA